

Symptom-based Diabetes Classification and Unsupervised Type Profiling

Filippo Velli
Digital Transformation Management
Università di Bologna

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Abstract

Driven by the critical clinical importance of the early detection of type 1 diabetes, this project proposes a multi-stage machine learning framework using a symptom-based clinical dataset to address the absence of explicit disease-type labels.

Utilising data from 520 patients collected at the Sylhet Diabetes Hospital, which tracks 16 categorical and numerical attributes (including age, gender, polyuria, polydipsia, and sudden weight loss) mapped to a binary target, the methodology establishes a dual-layer approach.

In the first stage, a supervised binary classification pipeline evaluates tabular classifiers—such as Decision Trees and Random Forests—to accurately distinguish between healthy and diabetic cohorts, mathematically prioritising classification over regression to target discrete diagnostic labels.

In the second stage, an unsupervised clustering layer utilising K-Means ($k=2$) is applied strictly to the predicted positive instances. This allows for an empirical evaluation of whether latent clinical signatures naturally partition into two distinct clusters aligning with known epidemiological profiles of type 1 (acute, early-onset) and type 2 diabetes.

To ensure methodological rigor, data preparation is managed via an end-to-end scikit-learn transformer pipeline to prevent data leakage

during categorical encoding and feature scaling, while stratified k-fold cross-validation and systematic hyperparameter tuning are implemented to mitigate overfitting risks on the 520-instance sample.

1 Introduction

Diabetes mellitus represents one of the most pervasive global health challenges of the modern era, characterised by chronic hyperglycemia resulting from defects in insulin secretion, insulin action, or both. Early and accurate detection of the condition is essential to mitigate severe long-term complications, including retinopathy, nephropathy, neuropathy, and cardiovascular disease. However, from a clinical management and epidemiological tracking perspective, simply diagnosing diabetes is insufficient; identifying the specific disease etiology—primarily distinguishing between Type 1 Diabetes Mellitus (T1DM), driven by autoimmune beta-cell destruction, and Type 2 Diabetes Mellitus (T2DM), driven by progressive insulin resistance—is critical for defining appropriate therapeutic pathways.

A primary challenge in utilising public clinical data for machine learning research is the frequent absence of explicit subtype labels. Public repositories, such as the symptom-based dataset collected from the Sylhet Diabetes Hospital, typ-

ically capture diagnostic outcomes as a simplistic binary classification task (Positive/Negative). While highly valuable for initial screening, this constraint leaves the underlying subpopulation structures hidden. This project addresses this explicit limitation by designing and implementing a robust dual-stage machine learning framework. Level 1 establishes a highly accurate supervised classifier to separate healthy individuals from diabetic patients. Level 2 subsequently isolates the predicted positive cohort and applies unsupervised learning to discover whether the underlying clinical signatures naturally partition into groups matching known T1DM and T2DM clinical proxies.

2 Proposed Method

The architectural design of the proposed multi-stage hybrid pipeline strictly follows the standardised CRISP-DM (Cross-Industry Standard Process for Data Mining) lifecycle. The framework is split into two primary computational layers: a Supervised Binary Classification layer (Level 1) and an Unsupervised Cohort Stratification layer (Level 2), bound together by a leakage-free pre-processing pipeline.

2.1 Data Pre-processing and Leakage Prevention

The dataset consists of clinical questionnaire records for 520 patients tracking 16 primary attributes: one continuous feature (*Age*), one demographic feature (*Gender*), and 14 binary clinical signs (e.g., *Polyuria*, *Polydipsia*, *Sudden weight loss*, *Obesity*) mapped to the target attribute *Class*.

To maintain clinical validity, this analysis deviates from traditional data cleaning practices, prioritising the preservation of patient diversity over standard statistical homogenisation:

Retention of Outliers: Statistical outlier removal (e.g., Z-scores or IQR) was omitted for the ‘Age’ attribute. Extremely young and old pa-

tients are retained as valid clinical profiles, preventing potential bias against atypical diagnostic cases.

Dataset Completeness: As the source dataset contained no null values across the 520 records, no imputation or deletion strategies were required.

Preservation of Categorical Diversity: Given that features are self-reported binary symptoms, variability in responses is interpreted as genuine clinical diversity rather than data noise. By treating this variance as phenotypic variation, we ensure the clustering phase remains sensitive to the underlying clinical signatures of individual patients.

To enforce absolute statistical integrity, a Stratified Train-Test Split (80/20 ratio) is executed before any pre-processing operations or parameter estimations occur. This ensures that validation matrices reflect true generalisation capabilities and completely eliminates data leakage. Following the split, a centralised **ColumnTransformer** handles the heterogeneous data types automatically:

1. **Categorical Encoding:** The binary textual features and gender flags are mapped to numeric spaces using a **OneHotEncoder** with the parameter `drop='first'` to avoid multicollinearity.
2. **Continuous Scaling:** Because the Level 2 clustering mechanism relies heavily on distance metrics, **MinMaxScaler** was intentionally selected to transform the *Age* attribute, scaling it strictly into the interval $[0, 1]$. This establishes absolute mathematical parity with the one-hot encoded binary features and prevents *Age* from dominating distance functions.

2.2 Level 1: Supervised Binary Classification

Level 1 focuses on separating healthy subjects from diabetic cases using tabular ensemble classifiers. The **Random Forest classifier** was

selected for the primary diagnostic screening phase. Its ensemble nature effectively manages complex, non-linear symptom interactions, offering greater stability and reliability than single decision trees, which are prone to overfitting. The interpretability provided by 'Feature Importance' metrics further supports clinical auditability, clarifying the symptomatic drivers of classification.

To optimise model convergence, hyperparameter optimisation is handled via a grid search engine combined with a 5-fold `StratifiedKFold` cross-validation strategy. The cross-validation framework optimises the model based on the F_1 -score, ensuring an equal balance between precision and recall across folds.

2.3 Level 2: Unsupervised Cohort Discovery

Once the Level 1 system achieves a robust validation profile, it processes the entire dataset. The framework extracts all instances predicted as *Positive* (diabetic) to form the independent Level 2 clustering cohort.

K-Means clustering was chosen for the type discovery. This choice was driven by:

- **Clinical Actionability:** Its centroid-based approach allows for the clear definition of clinical 'centers of gravity'. It was decided to use one approach focused on Type 1 Diabetes and another for Type 2 phenotypes.
- **Exhaustive Stratification:** K-Means guarantees that every patient is assigned a group, which is crucial for subsequent triage and the identification of 'Atypical/Mixed' cohorts through consensus analysis.
- **Binary Feature Stability:** It provides robust partitions in discrete binary feature spaces, where density-based algorithms like DBSCAN often struggle to produce interpretable results due to sensitivity to parameter settings.

The binary symptoms were treated in 2 ways: firstly, for both types of diabetes, all symptoms were treated equally to then apply a custom weighting function to emphasise "anchor symptoms" (e.g., Polyuria for Type 1, Obesity for Type 2). This modification to the distance matrix directs the K-Means algorithm to prioritise medically significant markers, effectively refining the cluster centroids to better align with established clinical phenotypes.

3 Results and Discussion

The experimental validation of the multi-stage pipeline produced clear quantitative metrics that validate both the classification and cohort discovery components.

3.1 Level 1 Classification Performance

The optimised Random Forest classifier achieved good generalisation metrics on the unseen test partition. The final parameters selected were `n_estimators=150`, `max_depth=8`, and `min_samples_split=5`. (Check table 1)

The model achieved an overall accuracy of 97%, with an F_1 -score of 0.98 for the positive class. The confusion matrix indicated that only 2 false negatives and 1 false positive were generated out of the 104 test instances.

To establish model interpretability and satisfy the requirement for explainable AI, feature importance analysis was conducted on the optimised Random Forest classifier. The analysis revealed that clinical symptoms—specifically *Polyuria* and *Polydipsia*—were the dominant predictors for diabetes screening, significantly outweighing demographic factors. Furthermore, an ablation experiment was performed by removing the *Age* and *Gender* features from the dataset. The model maintained its near-perfect accuracy without these demographic inputs, mathematically proving that the isolated clinical symptom profile is sufficient for robust

Level 1 screening.

3.2 Level 2 Clustering and Feature Selection Trade-offs

To analyze cluster quality, the silhouette profiles of distinct feature configurations within the isolated positive cohort ($N = 320$ predicted cases) were evaluated. The results reveal a clear mathematical trade-off between geometric cluster separation and clinical interpretability:

1. **Fewer Features Configuration (2 features: *Age, Sudden weight loss*):** Reached a high silhouette score of **0.823**. While geometrically compact, this configuration acts as a clinical trap; the algorithm creates clusters based almost entirely on an arbitrary age threshold, losing multidimensional diagnostic utility.
2. **Baseline Selected Configuration (4 features: *Age, Sudden weight loss, Obesity, Visual blurring*):** Produced a silhouette score of **0.422**. This performance represents the optimal balance for the project. Although the score is lower due to increased dimensionality, the clusters remain well-separated and capture highly meaningful clinical profiles.
3. **More Features Configuration (6 features: *Baseline + Irritability, Polyphagia*):** The silhouette score dropped to **0.246**. Adding features with high overlap across diabetic types introduces noise into the Euclidean distance calculations.

A critical methodological advancement in this phase was the implementation of a dual K -Means stability experiment to test targeted clinical hypotheses (evaluating Catabolic vs. Metabolic feature sets). In order to achieve this a feature weighting strategy was applied to the K -Means distance matrix to embed medical relevance into the clustering process. By

differentially weighting 'anchor symptoms' (e.g., Polyuria for Type 1, Obesity for Type 2), the algorithm is directed to prioritise medically significant markers, thereby enhancing the discovery of latent clinical signatures beyond standard Euclidean distance measures. By comparing independent clustering models, the framework calculated an Agreement Stability Score.

3.3 Impact of Feature Weighting on Phenotypic Discovery

To evaluate the efficacy of our feature-weighting strategy, the unweighted baseline was compared against a weighted framework. The results in Table 2 demonstrate that weighting anchor symptoms significantly improves phenotypic clarity. Notably, the weighted approach yielded a net gain of 8.46 percentage points in stable assignments, reducing the "Atypical/Mixed" cohort from 49.38% to 40.94%.

Furthermore, the increase in PC1 variance explained—rising from 36.51% to 51.82% in Approach 1—indicates that the model is capturing more structured clinical information rather than noise. This shift confirms that prioritising clinically significant indicators refines the cluster centroids, leading to more distinct and medically interpretable patient sub-groups.

The empirical properties of the discovered groups map closely onto established epidemiological definitions:

- **Cluster 0 (Type 1 Diabetes Proxy):** Characterised by a significantly younger average age (33.4 years) and a very high prevalence of acute sudden weight loss (84.6%), alongside a very low rate of obesity (4.1%). This aligns with the classic clinical presentation of T1DM.
- **Cluster 1 (Type 2 Diabetes Proxy):** Presents a much older demographic profile (mean age of 54.8 years), high rates of visual blurring (64.5%), a significantly higher prevalence of obesity (28.9%), and a lower

rate of sudden weight loss (31.2%). This matches the typical presentation of T2DM.

Other experimentations

The experimental phase included investigations into high-order ($k = 3$) clustering and composite feature indices. These methods were ultimately excluded from the primary dual-layer framework due to the "Consensus Paradox": while high-order initialisation improved geometric cohesion, it significantly increased diagnostic ambiguity, dropping consensus to approximately 31%. Furthermore, results indicate that the raw symptom data reaches a mathematical 'limit of resolution.' Forcing higher-order clustering without physiological tie-breakers (such as C-peptide levels or BMI) leads to the over-classification of 'Atypical' cases, which reduces the immediate clinical utility of the triage system. Despite their exclusion, these trials confirm that $k = 2$ remains the optimal configuration for current symptom-based datasets, with potential for future expansion as additional metabolic biomarkers are integrated.

4 Conclusion

This report successfully developed and validated a dual-stage machine learning framework to handle the lack of explicit disease subtype labels in public datasets. By leveraging a structured CRISP-DM approach, Level 1 established a highly accurate Random Forest classification pipeline (97% accuracy) that isolates diabetic patients without data leakage. Level 2 demonstrated that applying domain-driven feature selection and scaling enables K -Means clustering to successfully partition the diabetic cohort into two distinct subgroups serving as interpretable proxies for Type 1 and Type 2 diabetes.

While the current framework is highly effective based on the Sylhet symptom dataset, diagnostic subtyping would be significantly more robust with the inclusion of additional clinical indicators. Specifically, integrating data on HbA1c

levels, fasting plasma glucose, C-peptide levels (to measure endogenous insulin production), and body mass index (BMI) would allow for a more precise distinction between Type 1 and Type 2 presentations.

If these metrics were available, the "Atypical/Mixed" cohort would likely be reduced, as the model could better differentiate between metabolic insulin resistance (Type 2) and autoimmune-driven insulin deficiency (Type 1). Incorporating these physiological markers would bridge the gap between symptom-based triage and definitive clinical diagnosis.

References

- [1] Arsha Prasad (2021) *Diabetes Dataset*, Kaggle. (Accessed: 2026-07-13) <https://www.kaggle.com/datasets/arshaprasad/diabetes-dataset>.

Table 1: Level 1 Classification Performance Comparison

Model / Class	Precision	Recall	F_1 -Score	Support
<i>Decision Tree</i>				
Negative (Healthy)	0.98	1.00	0.99	40
Positive (Diabetic)	1.00	0.98	0.99	64
Accuracy			0.99	104
<i>Random Forest</i>				
Negative (Healthy)	0.97	0.97	0.96	40
Positive (Diabetic)	0.98	0.98	0.98	64
Accuracy			0.98	104

Table 2: Comparison of Clustering Performance: Unweighted vs. Weighted

Metric	Unweighted	Weighted	Change
Atypical/Mixed Cohort	49.38%	40.94%	-8.44%
Type 1-like Cohort	27.50%	38.75%	+11.25%
Type 2-like Cohort	23.12%	20.31%	-2.81%
Clustering Agreement	50.60%	59.06%	+8.46%
PC1 Variance (Appr. 1)	36.51%	51.82%	+15.31%
PC1 Variance (Appr. 2)	36.82%	45.87%	+9.05%